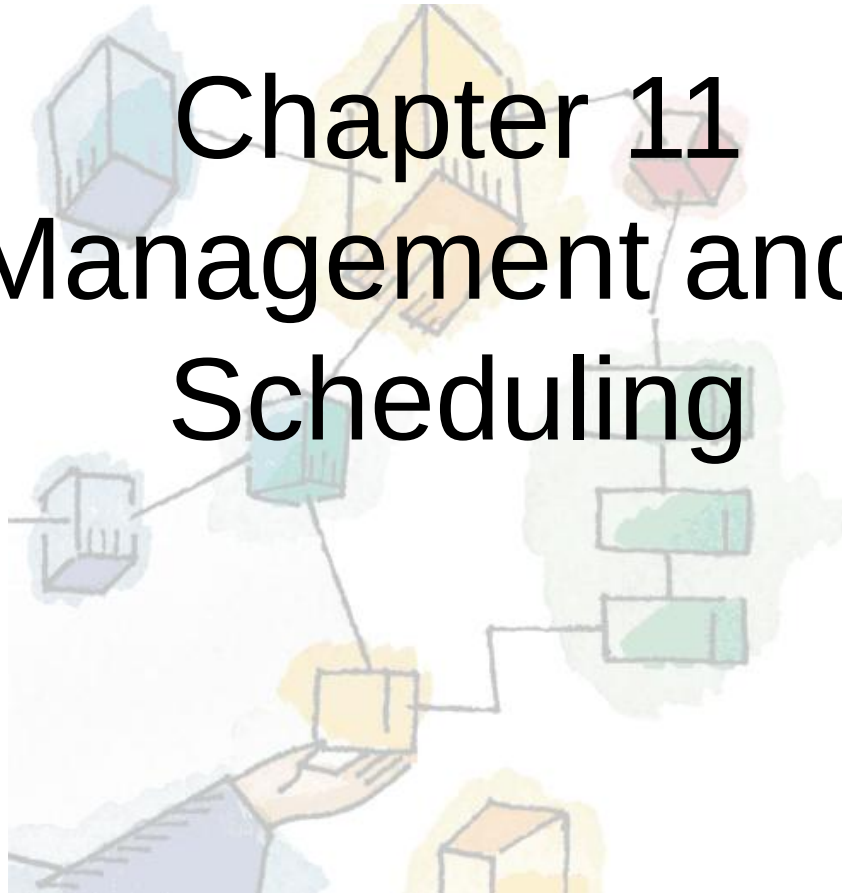
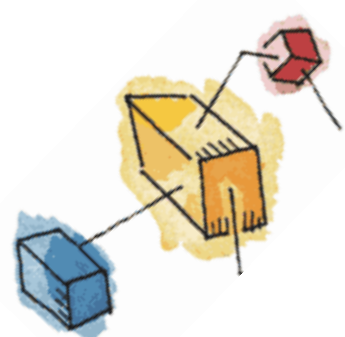


*Operating Systems:
Internals and Design Principles, 6/E*
William Stallings

Chapter 11

I/O Management and Disk Scheduling





I/O Management

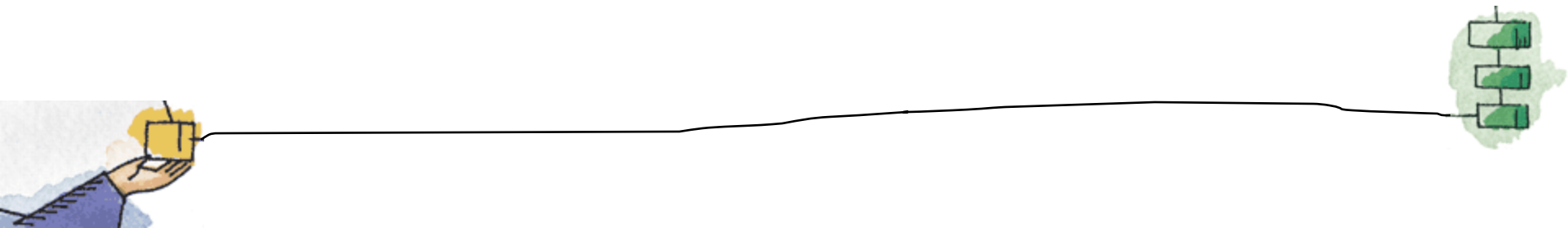
- Difficult area of OS design
 - Difficult to develop a consistent solution due to a wide variety of devices and applications
- Devices differ in a number of areas
 - Data Rate
 - Application
 - Complexity of Control
 - Unit of Transfer
 - Data Representation
 - Error Conditions

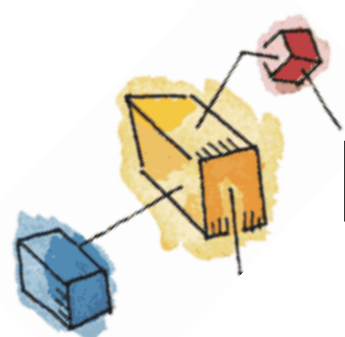




I/O Buffering

- Processes must wait for I/O to complete before proceeding
- It may be more efficient to perform input transfers in advance of requests being made and to perform output transfers some time after the request is made.





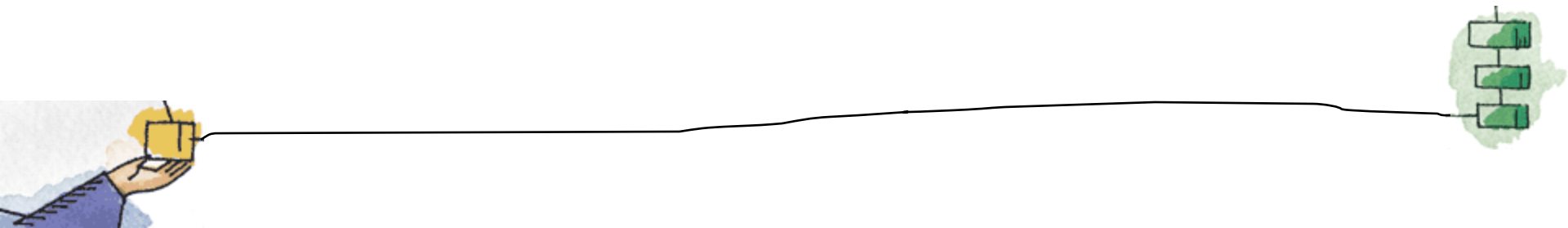
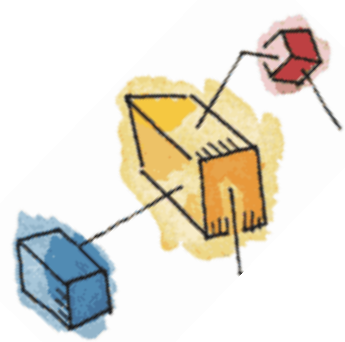
Block-oriented Buffering

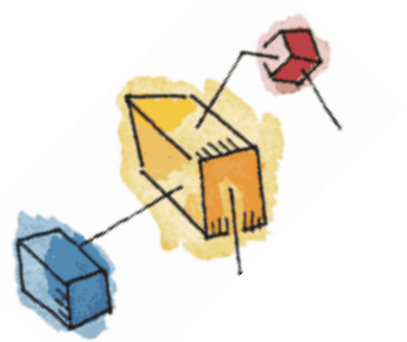
- Information is stored in **fixed sized blocks**
- **Transfers** are made a **block at a time**
 - Can reference data by block number
- Used for disks and USB



Stream-Oriented Buffering

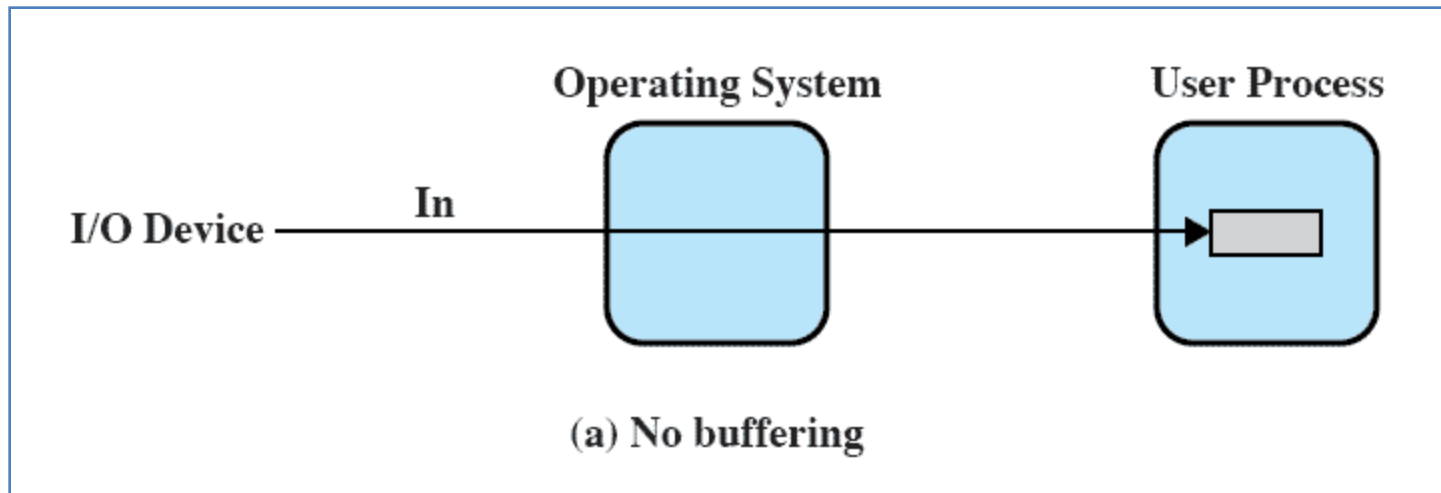
- Transfer information as a **stream of bytes**
- **Used for** terminals, printers, communication ports, mouse and other pointing devices, and most other devices that are not secondary storage

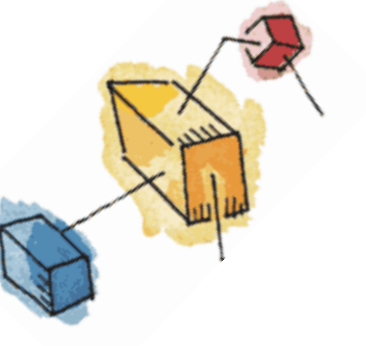




No Buffer

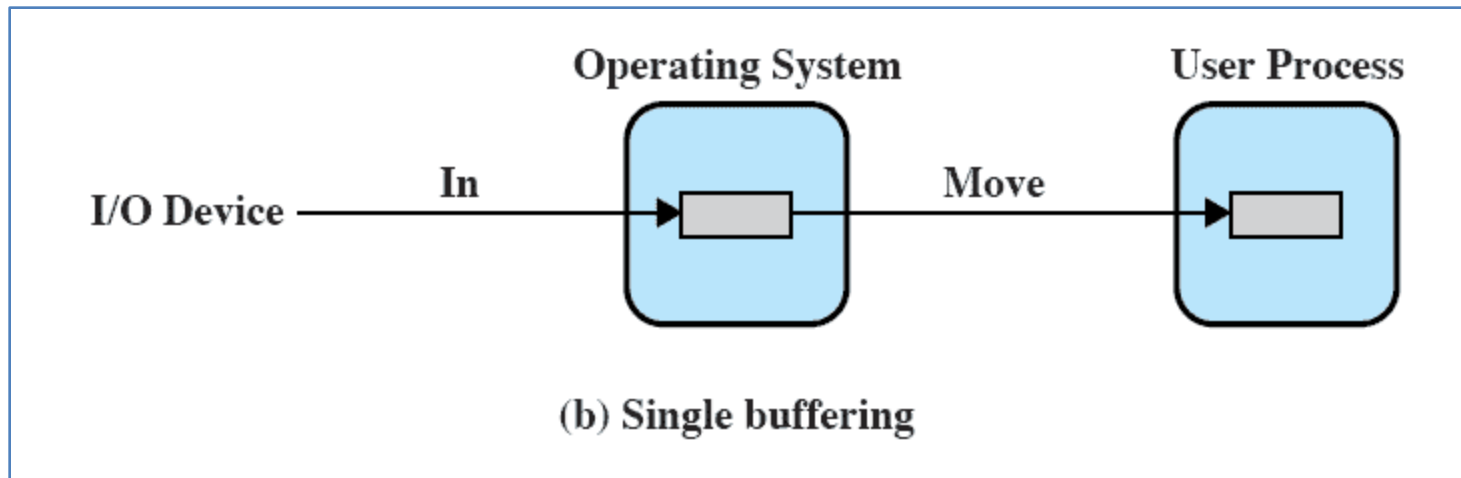
- Without a buffer, the OS directly access the device as and when it needs





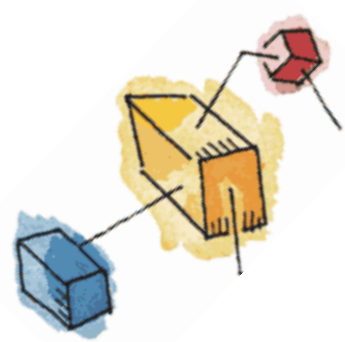
Single Buffer

- Operating system assigns **a buffer** in main memory for an I/O request



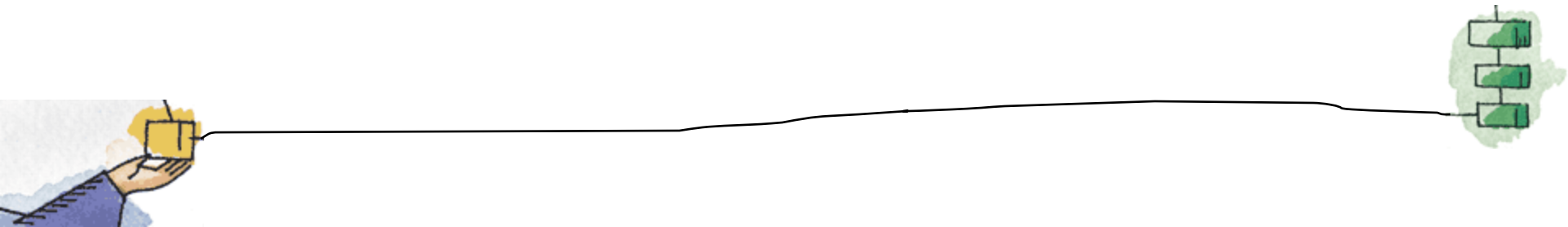
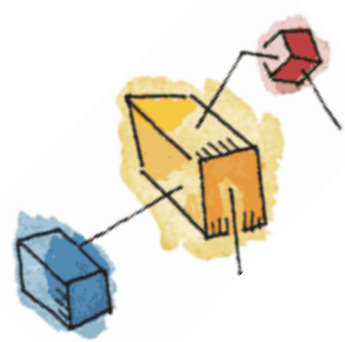
Block Oriented Single Buffer

- Input transfers made to buffer
- Block moved to user space when needed
- The next block is moved into the buffer
 - *Read ahead or Anticipated Input*
- Often a reasonable assumption as data is usually accessed sequentially



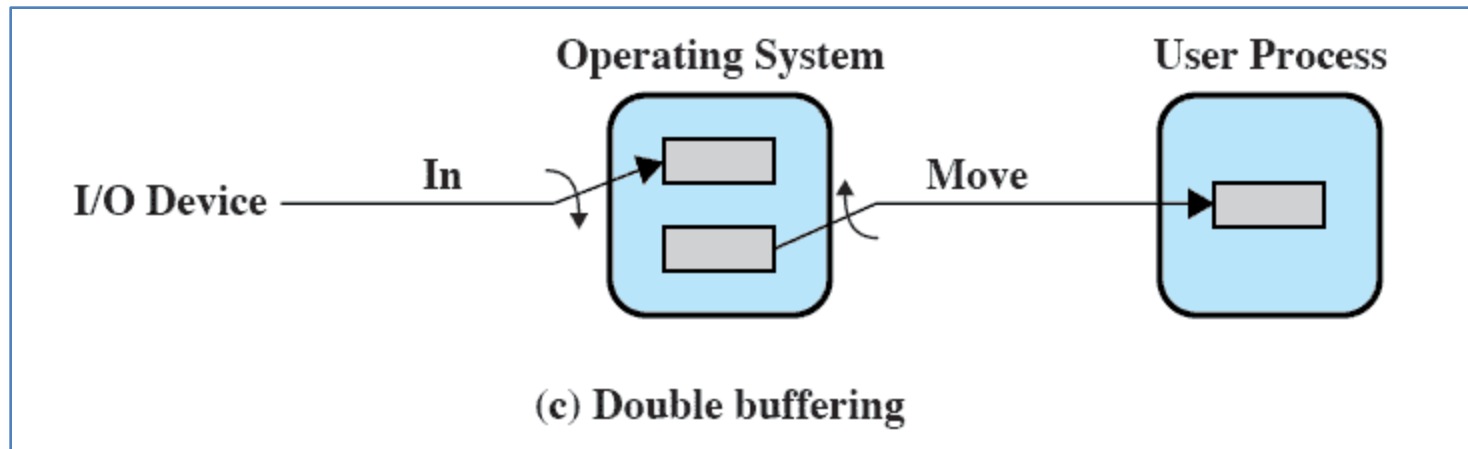
Stream-oriented Single Buffer

- Line-at-time or Byte-at-a-time
- **Terminals** often deal with **one line at a time** with carriage return signaling the end of the line
- **Byte-at-a-time** suites devices where a **single keystroke** may be significant
 - Also sensors and controllers



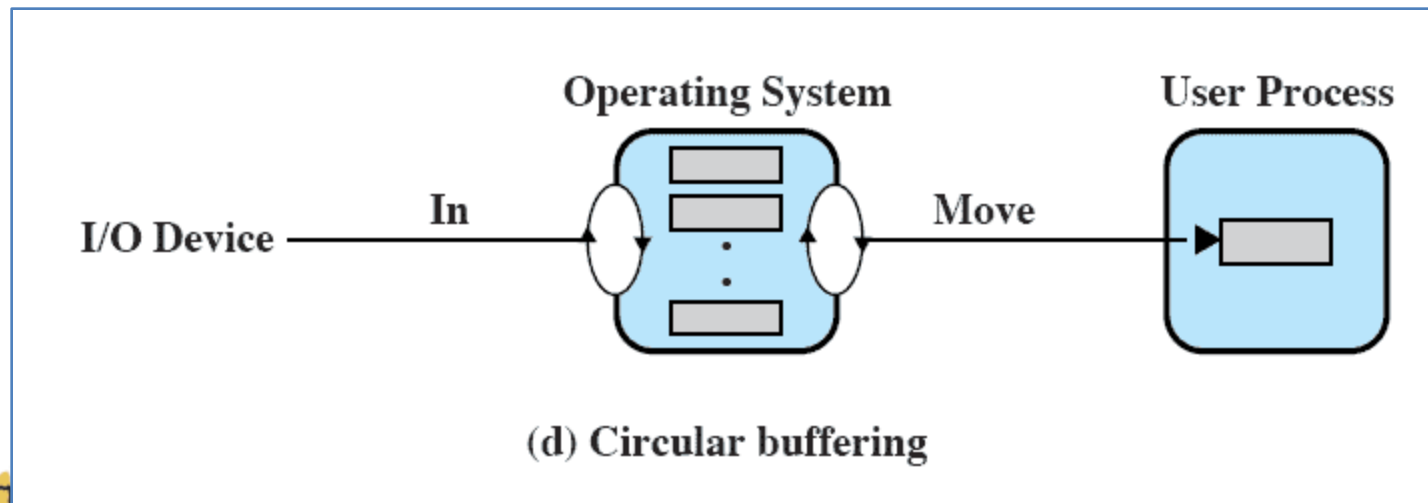
Double Buffer

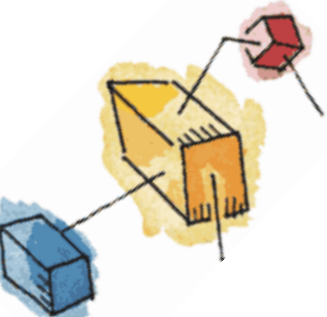
- Use **two system buffers** instead of one
- A process can transfer data to or from one buffer while the operating system empties or fills the other buffer



Circular Buffer

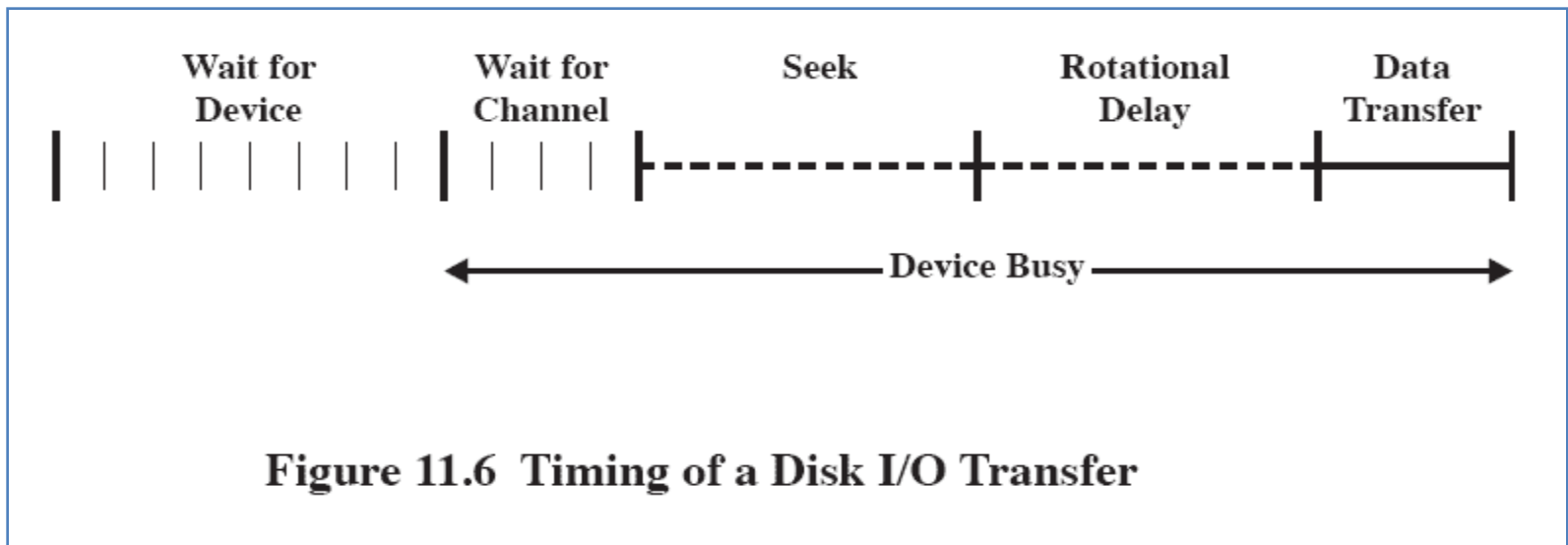
- More than two buffers are used
- Each individual buffer is one unit in a circular buffer
- Used when I/O operation must keep up with process

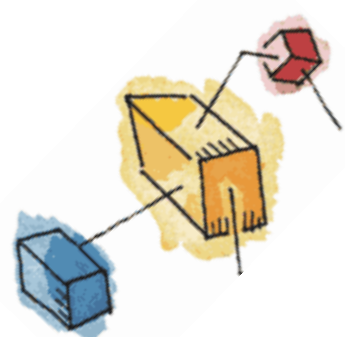




Disk Performance Parameters

- The actual details of disk I/O operation depend on many things
 - A general timing diagram of disk I/O transfer is shown here.





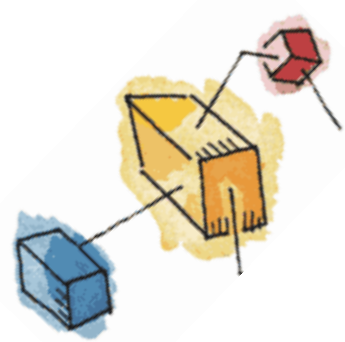
Positioning the Read/Write Heads

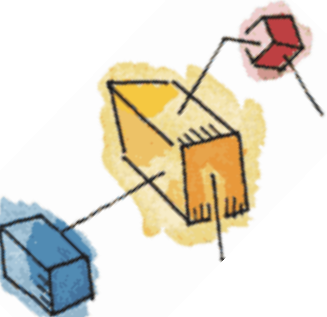
- When the disk drive is operating, the **disk is rotating** at constant speed.
- **Track selection** involves **moving the head** in a movable-head system **or** electronically **selecting one head** on a fixed-head system.



Disk Performance Parameters

- **Access Time** is the sum of:
 - **Seek time:** The time it takes to position the head at the desired track
 - **Rotational delay or rotational latency:** The time it takes for the beginning of the sector to reach the head
- **Transfer Time** is the time taken to transfer the data.

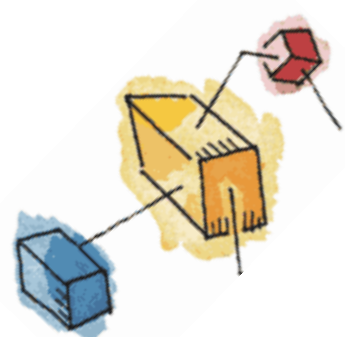




Disk Scheduling Policies

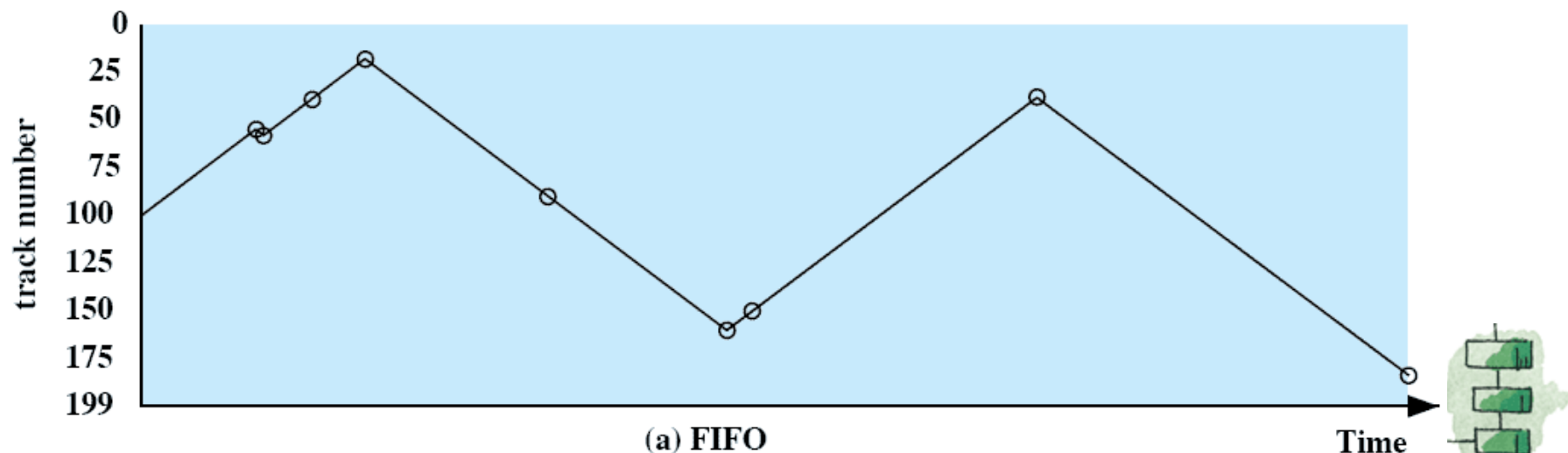
- To compare various schemes, consider a **disk head** is initially located at **track 100**.
 - Assume a **disk with 200 tracks** and that the disk request queue has random requests in it.
- The **requested tracks**, in the order received by the disk scheduler, are
 - 55, 58, 39, 18, 90, 160, 150, 38, 184.

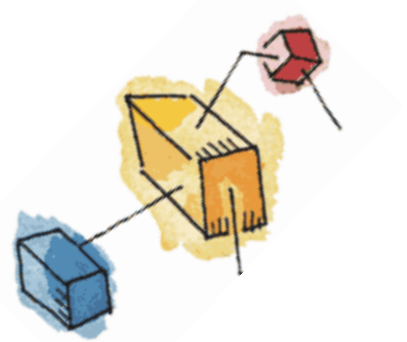




First-in, first-out (FIFO)

- 55, 58, 39, 18, 90, 160, 150, 38, 184
- Process request **sequentially**
- **Fair** to all processes, every request honored
- Can **become random scheduling** in performance if there are many processes competing for disk

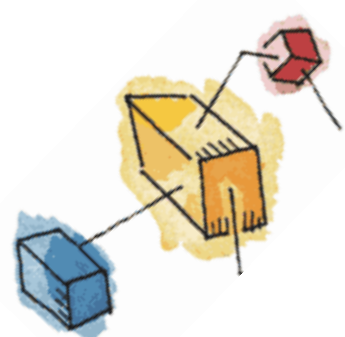




Priority

- **Goal** is not to optimize disk use but to meet other objectives
- **Short batch jobs** may have **higher priority**
- Provide **good interactive response time**
- **Longer jobs** may have to **wait** an excessively long time

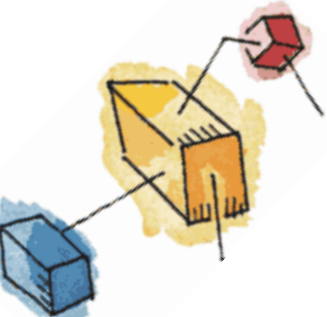




Last-in, first-out

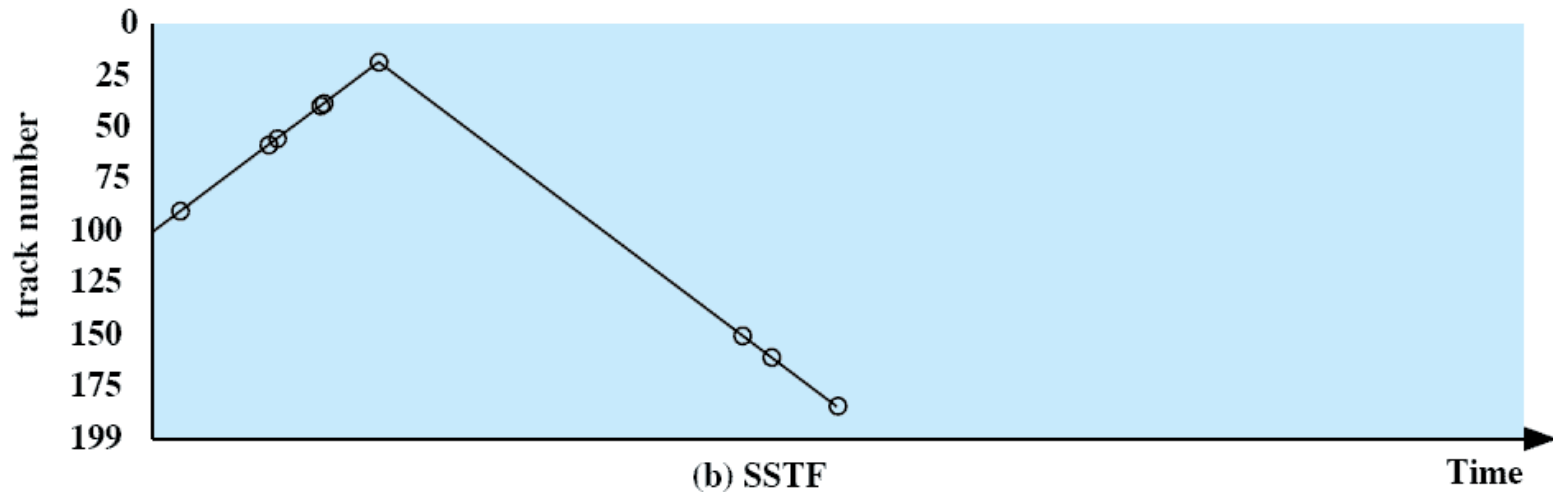
- Consider **most recent request** so there should be little arm movement (**Principle of locality**)
- Can provide **increased throughput** and **reduced queue length**
- **Possibility of starvation** since a job may never regain the head of the line





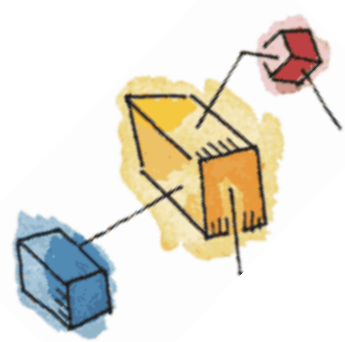
Shortest Service Time First

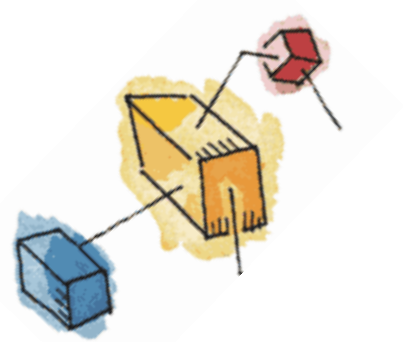
- Select the disk I/O request that requires the **least movement of the disk arm** from its **current position**
- Always choose the **minimum seek time**
- 55, 58, 39, 18, 90, 160, 150, 38, 184



SCAN

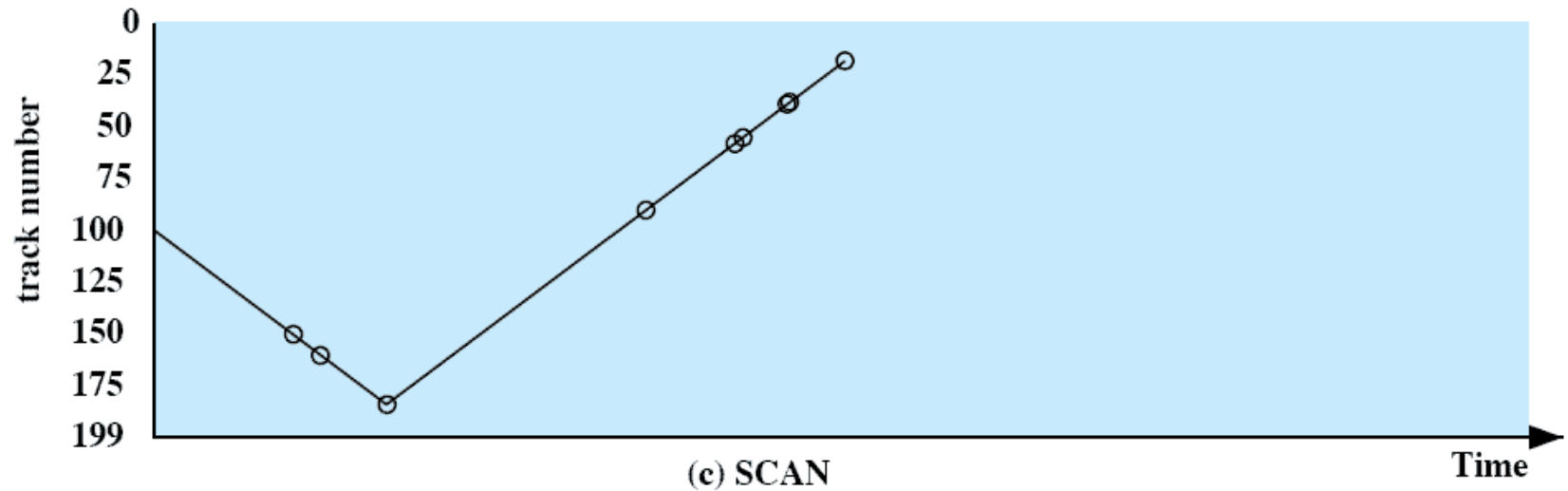
- Priority, LIFO and SSTF may leave some request unfulfilled, this is prevented by SCAN
- Arm moves in one direction only, satisfying all outstanding requests
 - until it reaches the last track in that direction or
 - there are no more requests in that direction
- then the direction is reversed
- Also known as the elevator algorithm



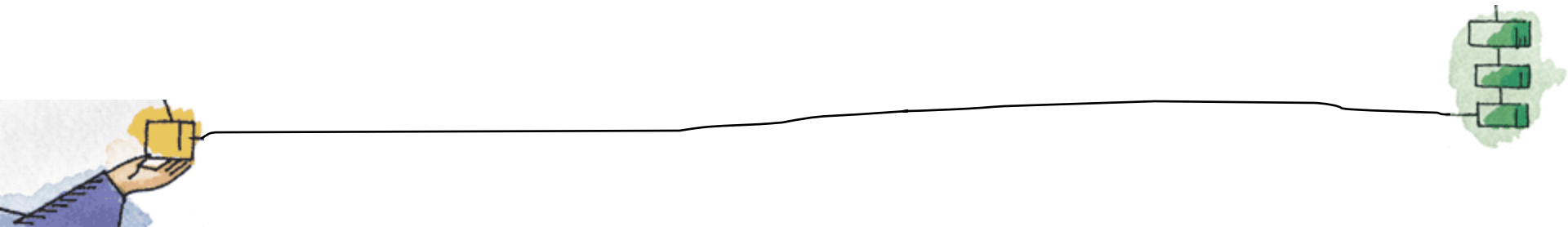


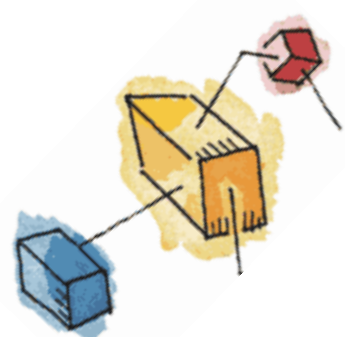
SCAN

- 55, 58, 39, 18, 90, 160, 150, 38, 184



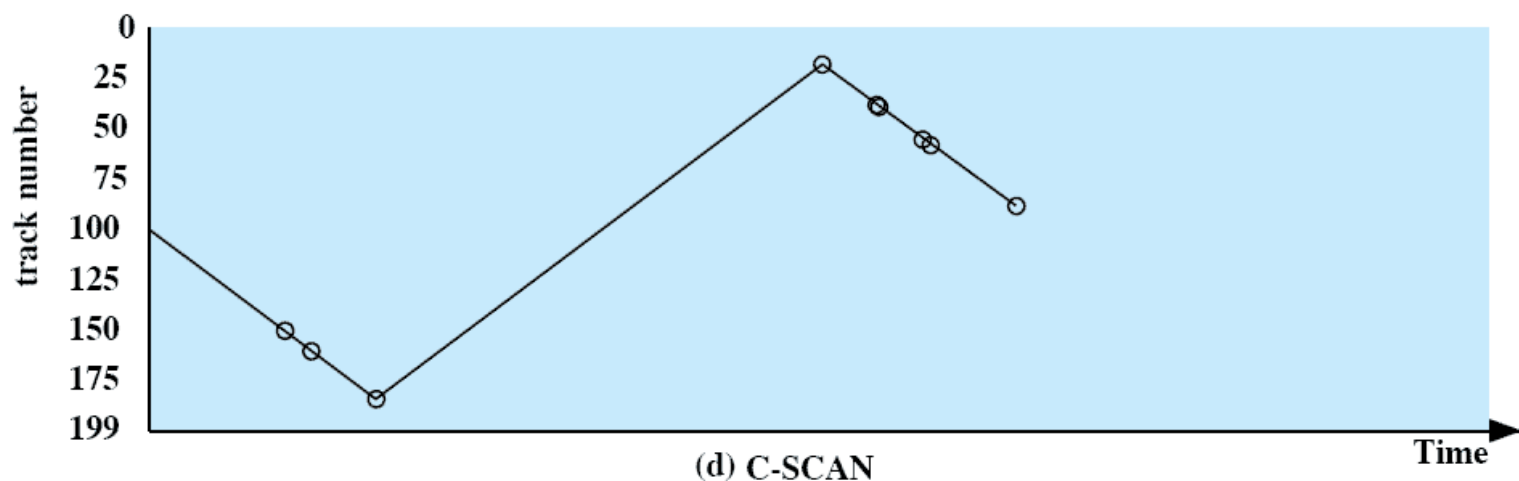
- SCAN is biased against the area most recently traversed





C-SCAN

- **Restricts** scanning to one direction only
- When the **last track has been visited** in one direction, the arm is **returned to the opposite end of the disk** and the **scan begins again**
- 55, 58, 39, 18, 90, 160, 150, 38, 184





Performance Compared

(a) FIFO (starting at track 100)		(b) SSTF (starting at track 100)		(c) SCAN (starting at track 100, in the direction of increasing track number)		(d) C-SCAN (starting at track 100, in the direction of increasing track number)	
Next track accessed	Number of tracks traversed	Next track accessed	Number of tracks traversed	Next track accessed	Number of tracks traversed	Next track accessed	Number of tracks traversed
55	45	90	10	150	50	150	50
58	3	58	32	160	10	160	10
39	19	55	3	184	24	184	24
18	21	39	16	90	94	18	166
90	72	38	1	58	32	38	20
160	70	18	20	55	3	39	1
150	10	150	132	39	16	55	16
38	112	160	10	38	1	58	3
184	146	184	24	18	20	90	32
Average seek length	55.3	Average seek length	27.5	Average seek length	27.8	Average seek length	35.8